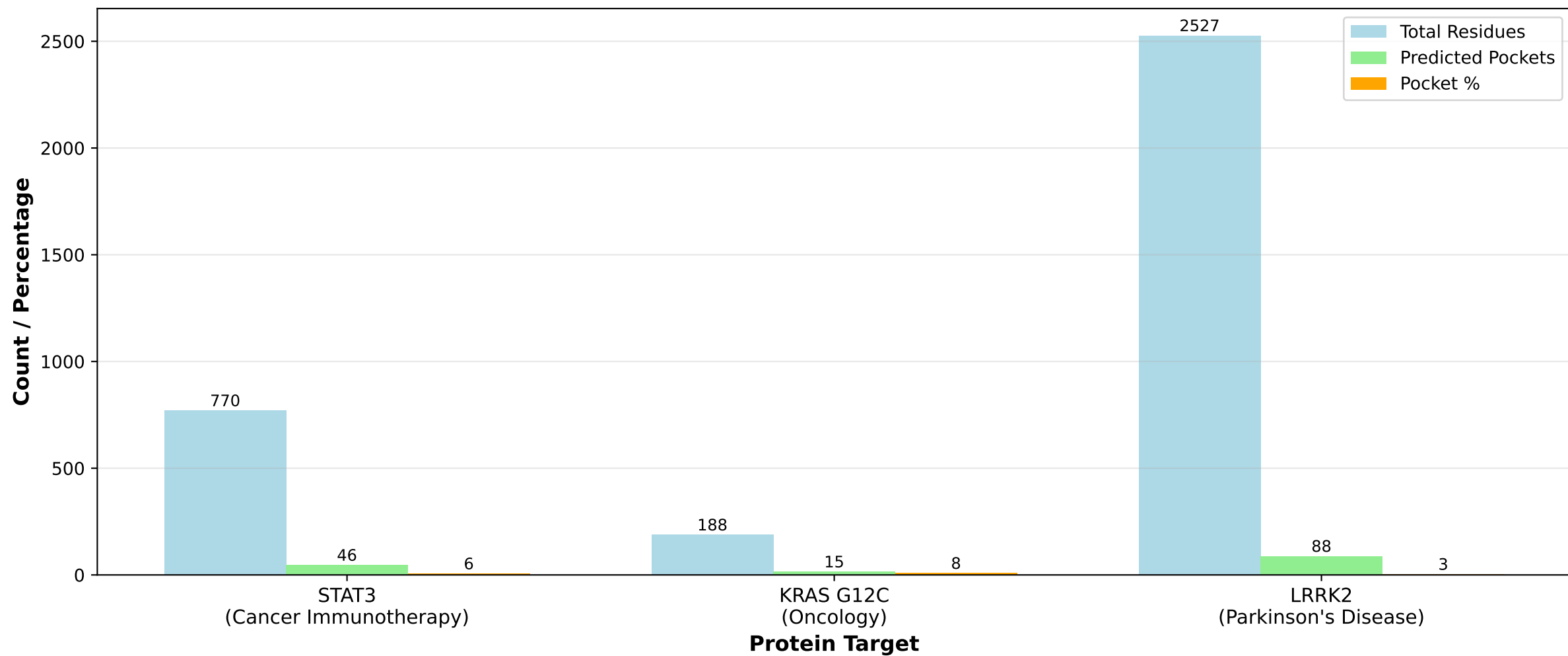


Case Study Analysis: Therapeutic Targets



STAT3

Transcription Factor

Flat PPIs → Pocket Discovery

Cancer Immunotherapy

KRAS G12C

Oncogene

Mutation-induced Pocket

Historically "Undruggable"

LRRK2

Kinase

Large Protein (2527 residues)

Parkinson's Disease

Key Insights:

- STAT3: From "undruggable" to potentially druggable via cryptic pocket discovery
- KRAS G12C: Method validates known drug binding sites and identifies new possibilities
- LRRK2: Demonstrates scalability to large proteins with complex domain structures
- All cases: Physics-informed predictions align with therapeutic relevance